University of Stuttgart
Collaborative
Artificial Intelligence

European Laboratory for Learning and Intelligent Systems

University of Stuttgart
Germany

Machine Perception and Learning

for Collaborative Intelligent Systems

Prof. Dr. Andreas Bulling

WS 2025/2026

Collaborative Artificial Intelligence Group, University of Stuttgart
www.collaborative-ai.org

Get to learn about our research and **possible open projects**!

- **When?** November 6th, 14:00
- **Where?** Pfaffenwaldring 5A, Room 0.009 (here)
- **What?** Research Intro + Mini Poster Session
- **Why?** Thesis spots are few and typically very competitive
- **More info?** Ilias and *`https://collaborative-ai.org`*
- **Questions?** constantin.ruhdorfer@vis.uni-stuttgart.de

Refresher Neural Networks

- Optimisation
- Activation Functions
- Weight Initialisation

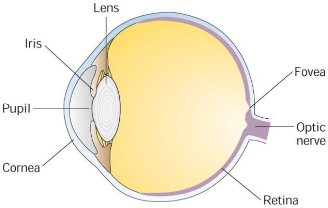
Convolutional Neural Networks (CNN)

- Human Visual Perception
- History of Deep Learning
- Deep Learning in a Nutshell
- Convolutional Neural Networks

Convolutional Neural Networks (CNN)

Human Visual Perception

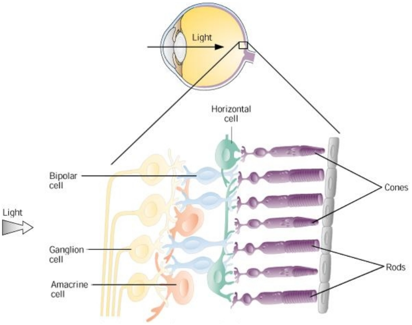
- Light passes through the **cornea** and **pupil**
  - Pupil adjusts light incidence
  - Focused on the **retina** by **lens**
  - Transduced by retinal photoreceptors



Source: Unknown

- **Fovea**: Dense array of photoreceptors
  - Receives light from objects being looked at directly
  - Area of highest visual acuity → *saccades*
- **Optic nerve** carries information on to the brain

- Hierarchy of cell layers
- **Receptor cells** (rods/cones) transduce light
- **Horizontal cells** convey information laterally
- **Bipolar cells** pool photoreceptors and connect to horizontal cells



Source: Unknown

- **Amacrine cells** modulate signals
- **Ganglion cells** form the optic nerve

Photoreceptor cells

- Differ in sensitivity, number, location, response time, and wavelength

Retinal ganglion cells

- Different populations
  - M, P, and K cells
  - Intrinsic photosensitive cells
- Different response to contrast, color, shape, texture, and motion

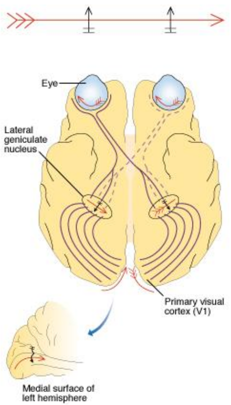
- **Rods**: Night vision
- **Cones**: Colour vision and fine details

#	Rods	Cones
Sensitivity	+	-
Number	90×10^6	4.5×10^6
Location	periphery	center
$T_{Response}$	-	+
Wavelength	+	-
#Pigments	1	3
#Photons	1	10-100

Optic nerves terminate in two **lateral geniculate nuclei (LGN)**

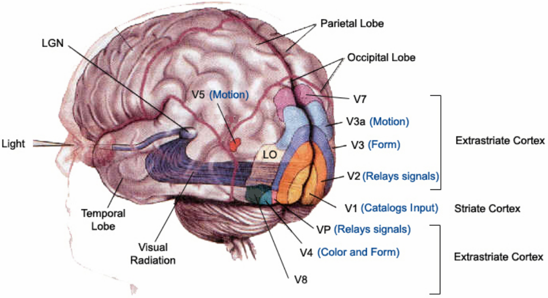
- Part of the central nervous system
- Function mostly unknown
  $\rightarrow$ Temporal de-correlation?

  [Dong and Atick, 1995]
- Segregates visual information into physically distinct pathways
- Act as "relay station" between retina and the visual cortex



Source: Unknown

- Largest system in the brain: 40×10^6 neurons
- Active area of research, not fully understood

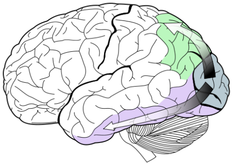


Source: Unknown

- Cortical hierarchy:
  V1 (striate/primary visual cortex)
  V2-V8 (secondary visual areas)

Two-streams hypothesis [Goodale and Milner, 1992; Norman, 2002]

Dorsal stream (V1 → V2 → V5 → V6)



Source: [Jenkins, 2010]

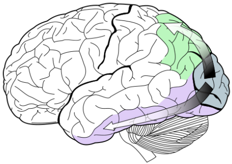
- **Dorsal stream**: the ”where pathway”
  - Visually guided action (e.g. eyes, head, limbs)

Two-streams hypothesis [Goodale and Milner, 1992; Norman, 2002]

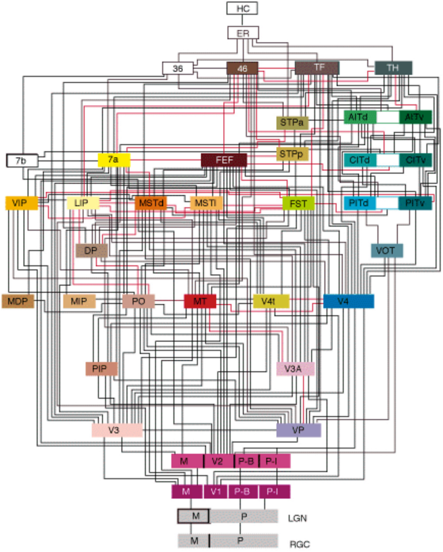
Dorsal stream (V1 → V2 → V5 → V6)

Ventral stream (V1 → V2 → V4 → IT)

- **Dorsal stream**: the "where pathway"
  - Visually guided action (e.g. eyes, head, limbs)
- **Ventral stream**: the "what pathway"
  - Representation of the visual world
  - Visual memory
  - Object identification and recognition



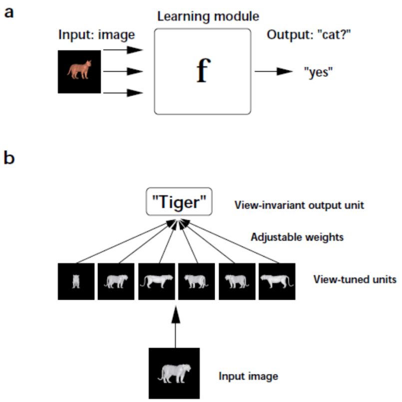
Source: [Jenkins, 2010]



Source: [Felleman and Van Essen, 1991]

Goal: Good classification performance

- Trade-off between
  - Specificity
  - Invariance
    - Affine transformations
    - Lighting
- Impacts generalisation ability



Source: [Riesenhuber and Poggio, 2000]

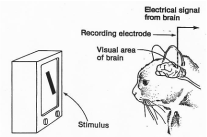
1959

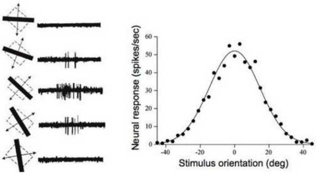
RECEPTIVE FIELDS OF SINGLE NEURONES IN THE CAT'S STRIATE CORTEX

1962

RECEPTIVE FIELDS, BINOCULAR INTERACTION AND FUNCTIONAL ARCHITECTURE IN THE CAT'S VISUAL CORTEX

1968...

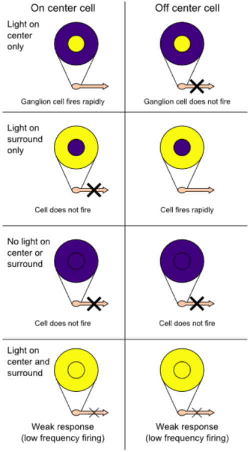




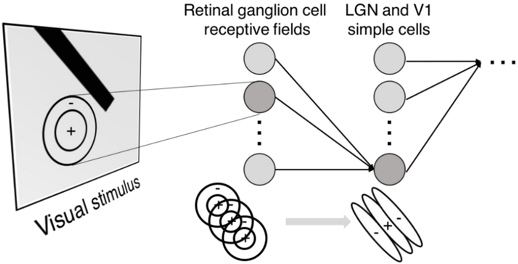
Source: [Hubel and Wiesel, 1959, 1962]

https://www.youtube.com/watch?v=8VdFf3egwfg

- All receptors synapsing with a particular cell collectively form its **receptive field**
    - E.g., a group of ganglion cells in the retina form the receptive field of a cell in the brain
- Excitatory / inhibitory regions → on-center / off-center cells
- Sensitive to contrast + orientation of a bar, edge, or gratings



Source: original upload by delldot, modified by Xoneca, Wikimedia Commons



Simple cells:
Response to light orientation

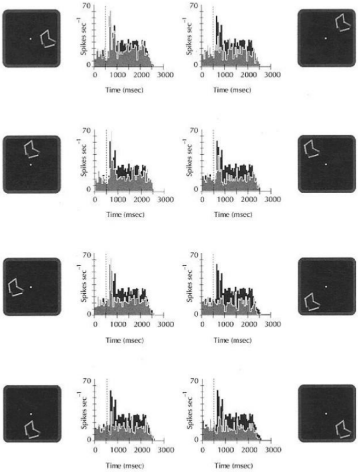
Complex cells:
Response to light orientation and movement

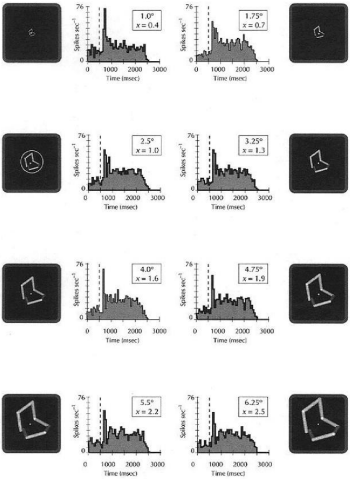
Hypercomplex cells:
response to movement with an end point

No response

Response (end point)

Source: Lane McIntosh, Stanford University



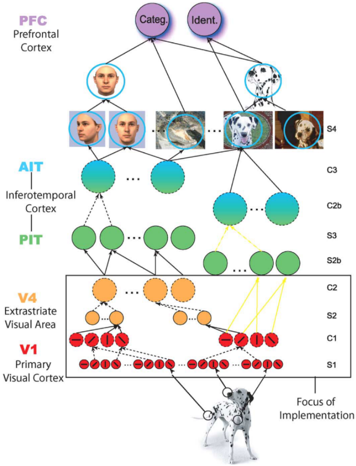


Source: [Logothetis et al., 1995]

Convolutional Neural Networks (CNN)

History of Deep Learning

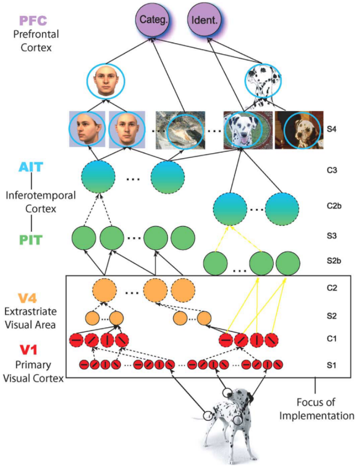
- ”Hierarchical Model and X”
- Models ”immediate object recognition” process
- Covers the first couple of hundreds of milliseconds
- Before any top-down influences, such as attention shifts or eye movements



Source: [Folowosele et al., 2011]

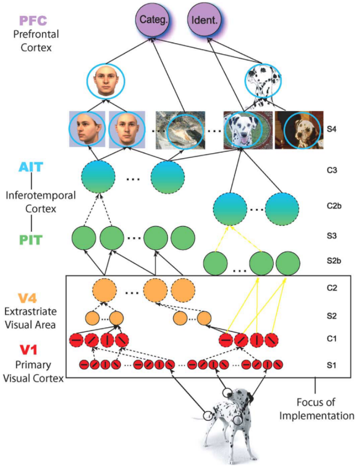
esenhuber and Poggio, 2000; Serre et al., 2007]

- S (simple) cells: tuned to specific stimuli, typically small receptive fields



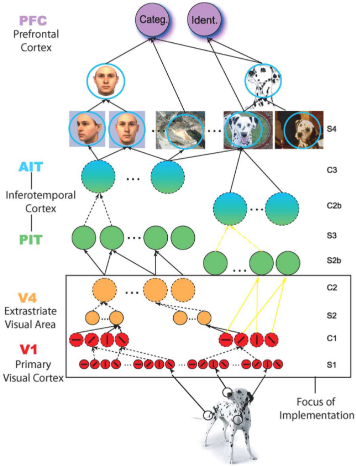
Source: [Folowosele et al., 2011]

- S (simple) cells: tuned to specific stimuli, typically small receptive fields
- C (complex) cells: combine output from multiple S units to increase invariance and receptive field



Source: [Folowosele et al., 2011]

- **S** (simple) cells: tuned to specific stimuli, typically small receptive fields
- **C** (complex) cells: combine output from multiple S units to increase invariance and receptive field
- Many iterations of these operations allow to construct complex objects from low-level features

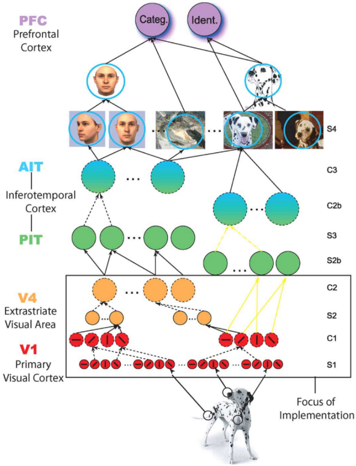


Source: [Folowosele et al., 2011]

- S-cell response to synaptic inputs $(x1, \ldots, x_{n_{S_k}})$ from the previous layer:

  $$y = exp(-\frac{1}{2\sigma^2}\sum_{j=1}^{n_{S_k}}(w_j - x_j)^2)$$

- σ defines the sharpness of the bell-shaped tuning operation
- **Trainable parameters** w

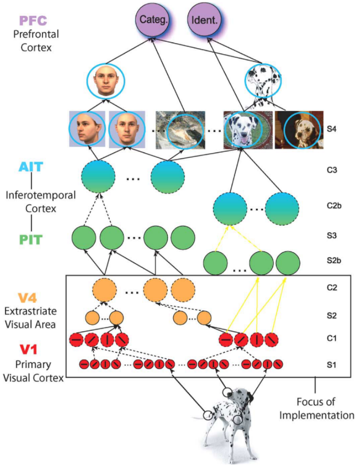


Source: [Folowosele et al., 2011]

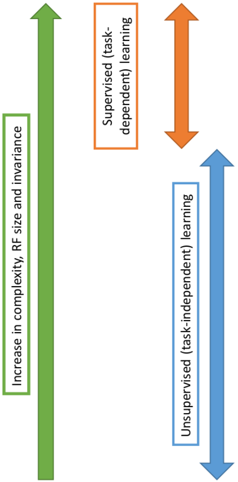
- C-cell response to synaptic inputs $(x1, \ldots, x_{n_{C_k}})$ from the previous layer:

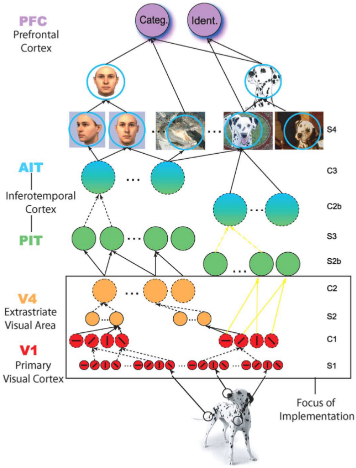
  $$y = \max_{j=1 \ldots n_{C_k}} x_j$$

- Response corresponds to the response of the strongest of its afferents $(x1, \ldots, x_{n_{C_k}})$
- **Pooling** operation

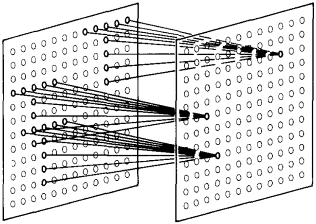


Source: [Folowosele et al., 2011]

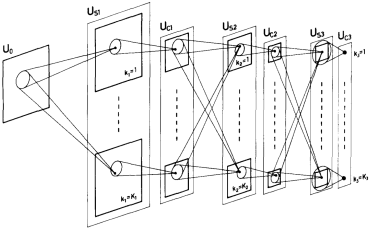




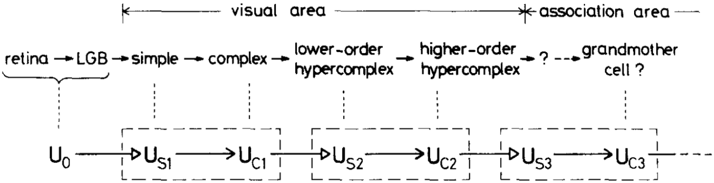
Source: [Folowosele et al., 2011]



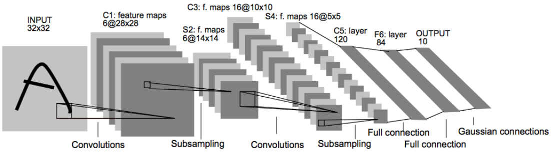
Source: [Fukushima and Miyake, 1982]



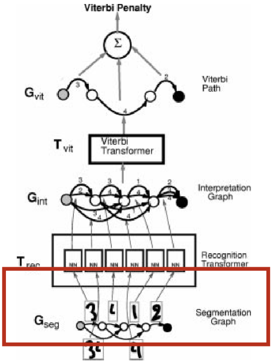
Source: [Fukushima and Miyake, 1982]



Source: [Fukushima and Miyake, 1982]

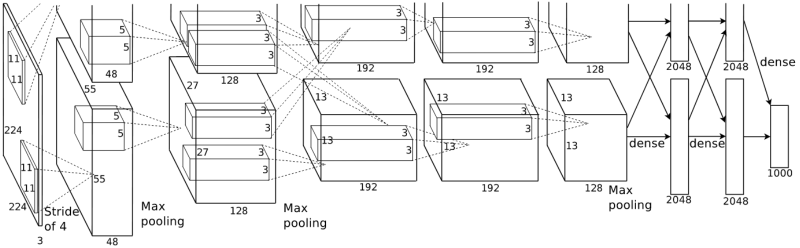


Source: [LeCun et al., 1998]



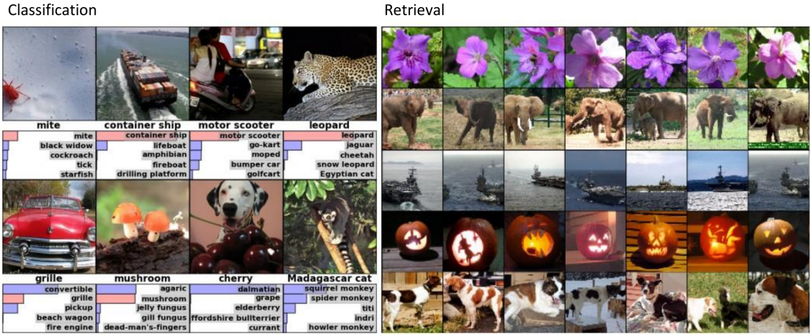
Source: [LeCun et al., 1998]

Around **60,000 parameters**



Source: [Krizhevsky et al., 2012]

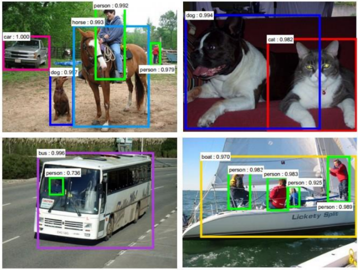
Around **60.000.000 parameters**, trained on two GPUs



Source: [Krizhevsky et al., 2012]

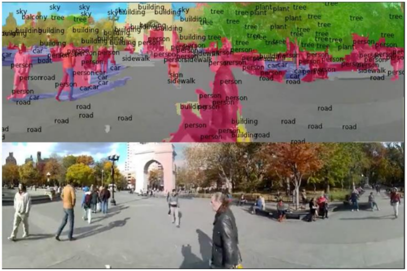
Evaluated on the large-scale ImageNet dataset [Deng et al., 2009]

Detection



Source: [Ren et al., 2015]

Segmentation

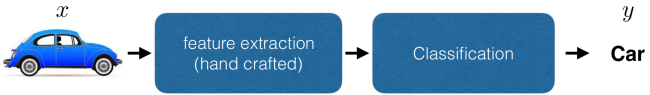


Source: [Farabet et al., 2012]

Further related work: [Girshick, 2015; He et al., 2017]

Convolutional Neural Networks (CNN)

Deep Learning in a Nutshell



Source: Bernt Schiele, MPII

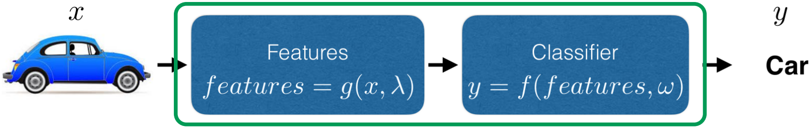
- Extraction of image features
    - Often handcrafted and fixed
    - Often too general (not task-specific enough)
    - Often too specific (do not generalise well to other tasks)
- How to achieve the best classification performance?
    - More complex classifier (e.g. multi-feature, non-linear)?
    - How specialised for the task?

x

→ **Features** $features = g(x, \lambda)$ → **Classifier** $y = f(features, \omega)$ → y **Car**

Source: Bernt Schiele, MPII

- Parametrised feature extraction
- Features should be
  - Efficient to compute
  - Efficient to train (differentiable)



Source: Bernt Schiele, MPII

”end-to-end system”

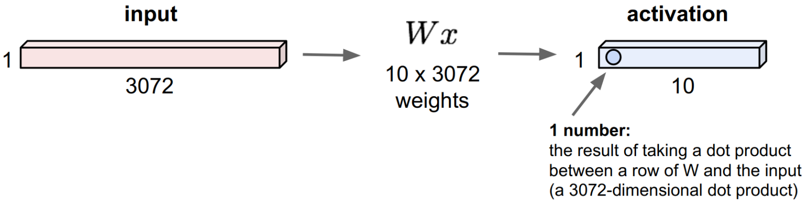
- Parametrised feature extraction
- Features should be
  - Efficient to compute
  - Efficient to train (differentiable)
- **Joint** training of **feature** extraction and **classification**

1. **Learning of features** (across many layers)
2. **Efficient** and **trainable** systems by **differentiable** building blocks
3. Composition of deep architectures via **non-linear modules**
4. "**End-to-end**" training: no differentiation between feature extraction and classification

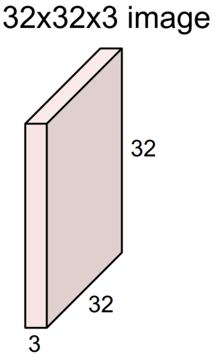
Convolutional Neural Networks (CNN)

Convolutional Neural Networks

$32 \times 32 \times 3$ image $\rightarrow$ flatten to 3072×1



Source: Fei-Fei Li, Stanford University



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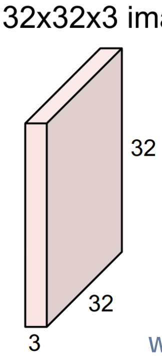
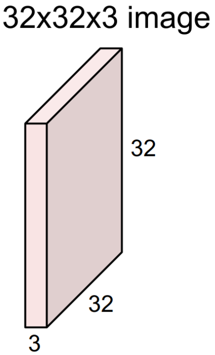


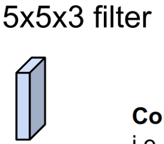
height

width

Source: Fei-Fei Li, Stanford University

depth

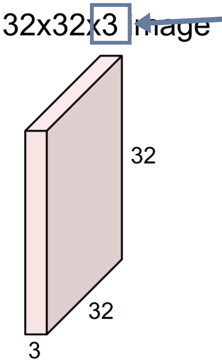


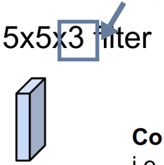


Convolve the filter with the image
i.e. “slide over the image spatially, computing dot products”

Source: Fei-Fei Li, Stanford University

Filters always extend the full depth of the input volume





Convolve the filter with the image
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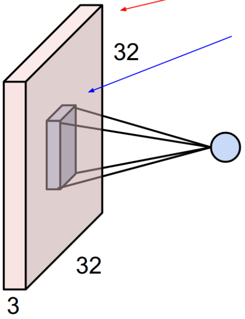
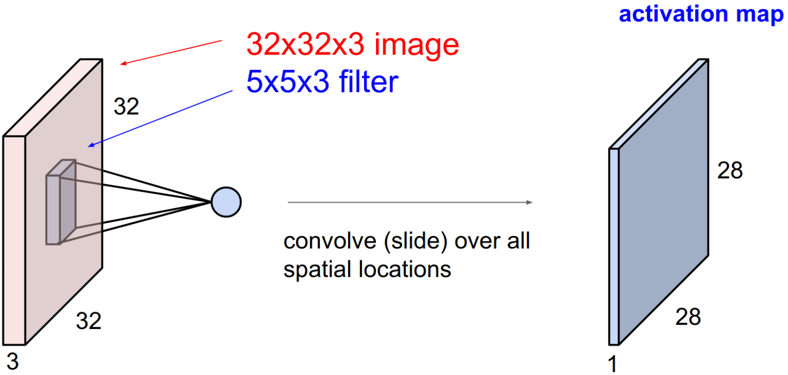


1 number:
the result of taking a dot product between the filter and a small 5x5x3 chunk of the image
(i.e. 5*5*3 = 75-dimensional dot product + bias)

$$w^T x + b$$

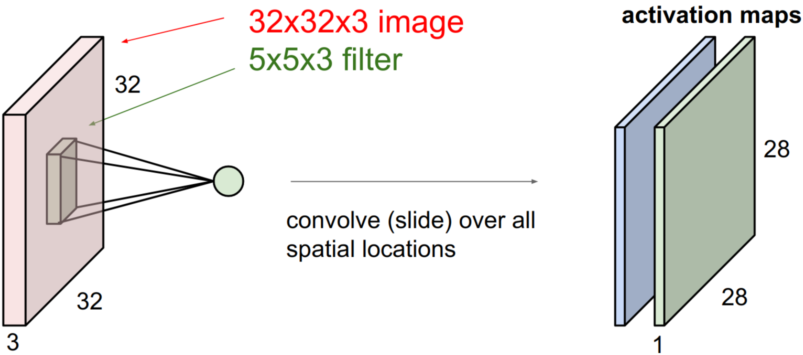
Source: Fei-Fei Li, Stanford University



Source: Fei-Fei Li, Stanford University

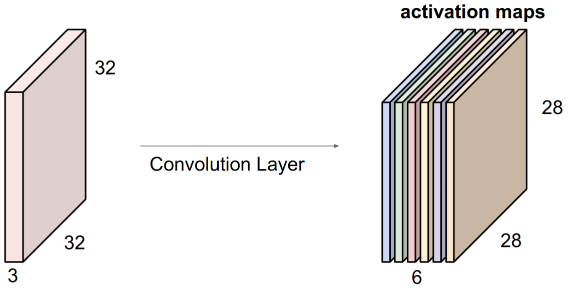
also known as “**feature map**”

Consider a second, green filter:



Source: Fei-Fei Li, Stanford University

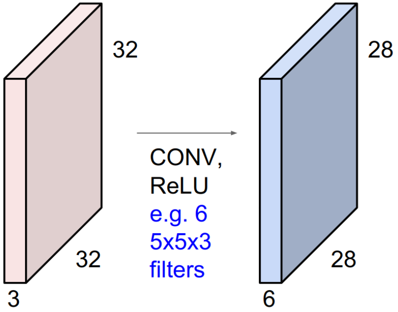
For example, if we had **six** 5×5 **filters**, we'll get six separate activation maps:



Source: Fei-Fei Li, Stanford University

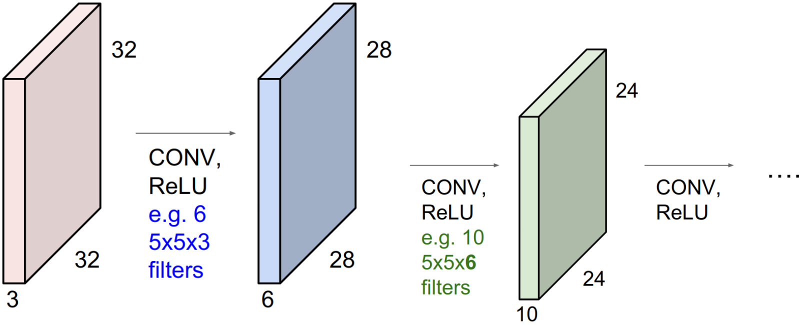
Stacking them up results in a "new image" of size $28 \times 28 \times 6$

CNNs are a sequence of convolutional layers interspersed with activation functions



Source: Fei-Fei Li, Stanford University

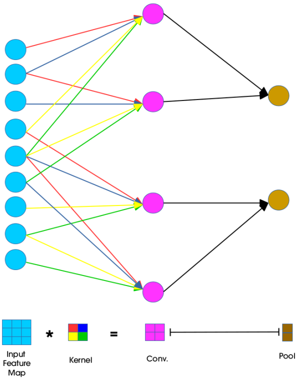
CNNs are a sequence of convolutional layers interspersed with activation functions



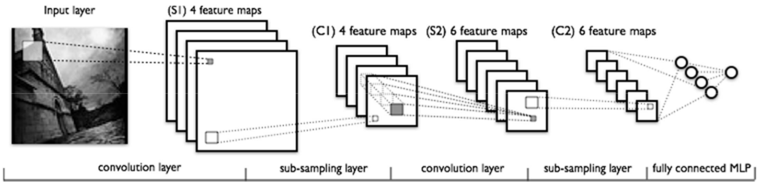
Source: Fei-Fei Li, Stanford University

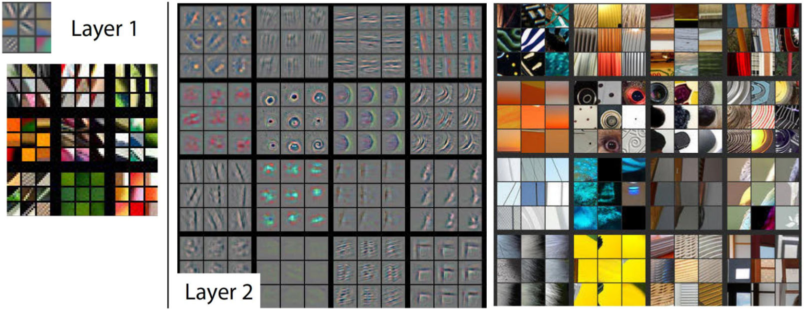
Advantages

- Reduces the number of weights that must be learned
- Reduces model training time
- Makes feature search insensitive to feature location

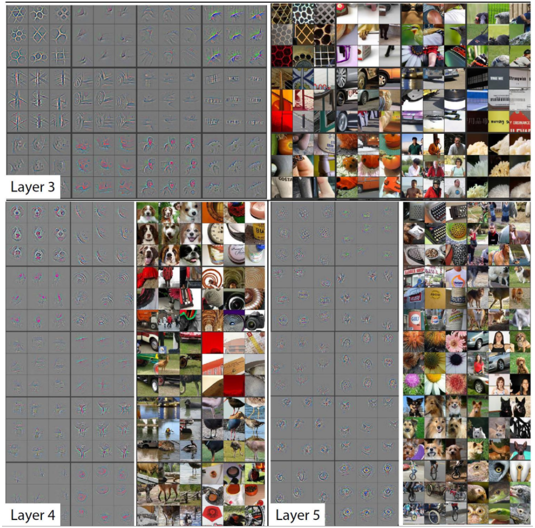


Source: www.jefkine.com

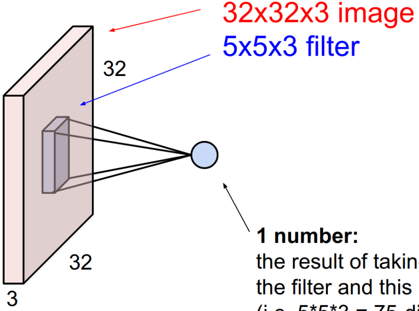


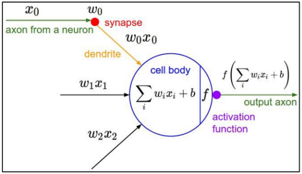


Source: [Zeiler and Fergus, 2014]



Source: [Zeiler and Fergus, 2014]



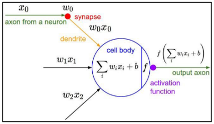


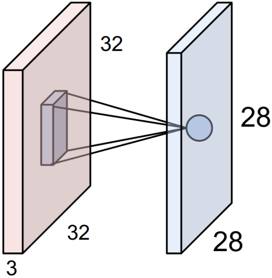
It's just a neuron with local connectivity...

1 number:
the result of taking a dot product between the filter and this part of the image
(i.e. 5*5*3 = 75-dimensional dot product)

Source: Fei-Fei Li, Stanford University

Convolutional Neural Networks – Brain/Neuron view



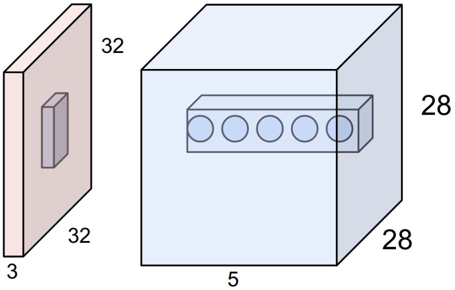


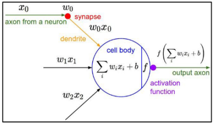
An activation map is a 28x28 sheet of neuron outputs:

1. Each is connected to a small region in the input
2. All of them share parameters

"5x5 filter" -> "5x5 receptive field for each neuron"

Source: Fei-Fei Li, Stanford University



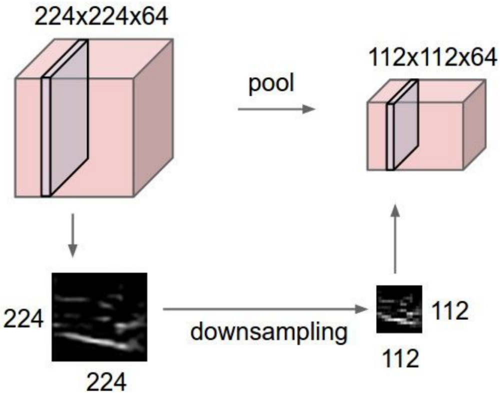


E.g. with 5 filters, CONV layer consists of neurons arranged in a 3D grid (28x28x5)

There will be 5 different neurons all looking at the same region in the input volume

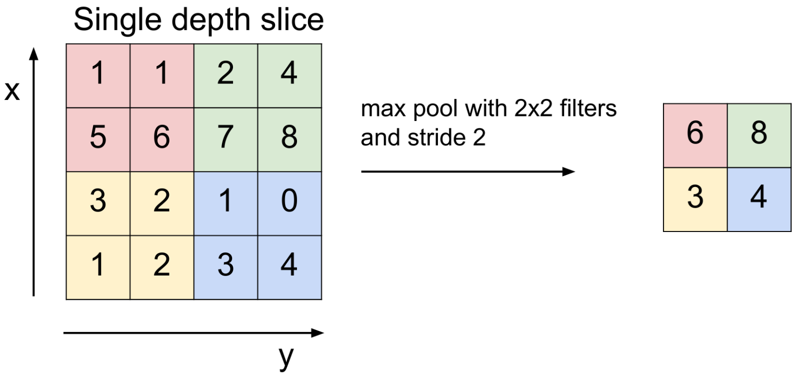
Source: Fei-Fei Li, Stanford University

- Makes the representations smaller and more manageable
- Operates over each activation map independently

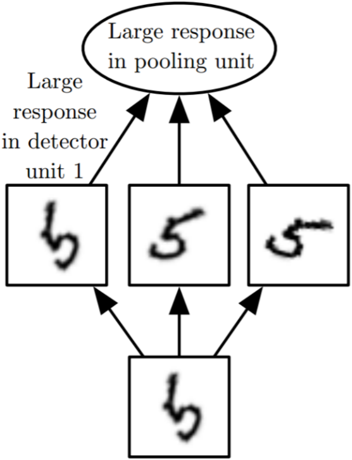


Source: Fei-Fei Li, Stanford University

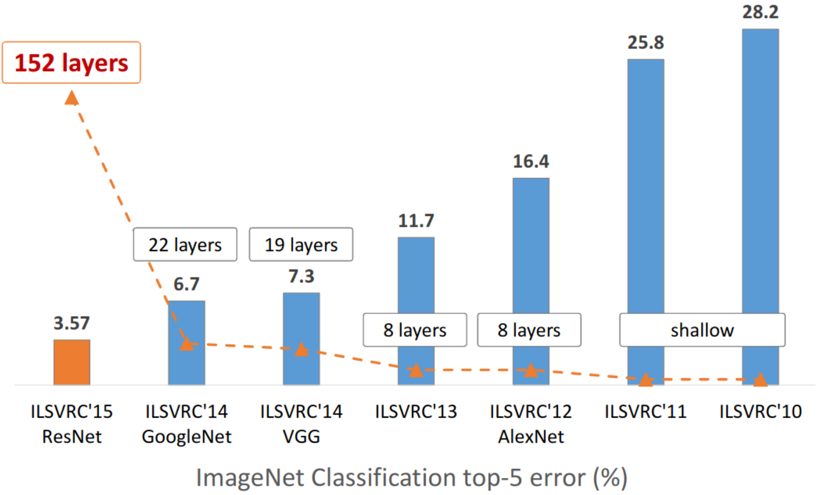
Convolutional Neural Networks – Max Pooling



Source: Fei-Fei Li, Stanford University

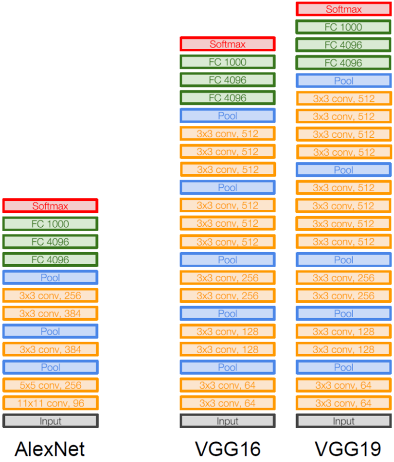


Source: www.deeplearningbook.org

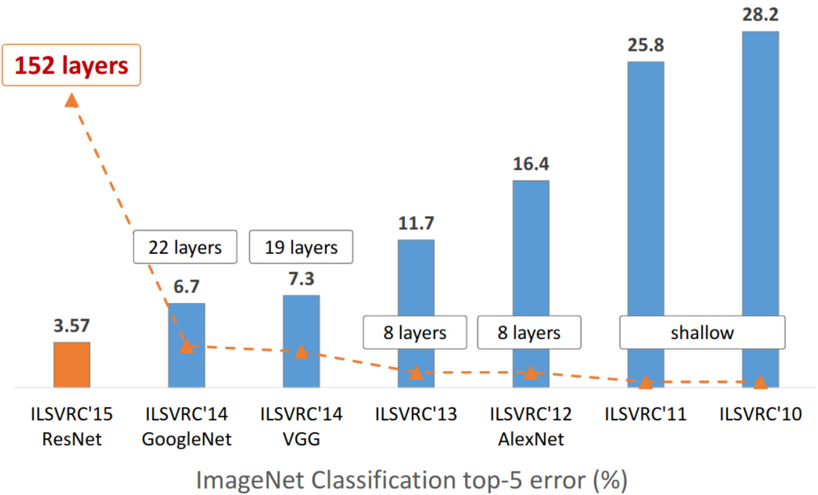


Source: Kaiming He

- Smaller filters
- More layers
- Due to depth:
    - Large receptive field
    - Despite smaller filters
- Due to smaller filters:
    - Fewer parameters

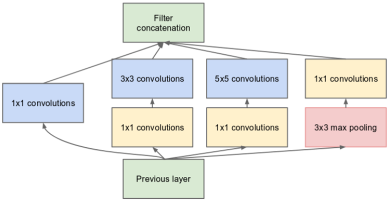


Source: [Simonyan and Zisserman, 2014]

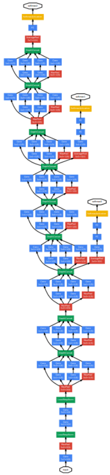


Source: Kaiming He

- 22 layers
- Efficient "Inception" module – "network within network"
- No fully-connected layers
- Only five million parameters
  - 12x less than AlexNet

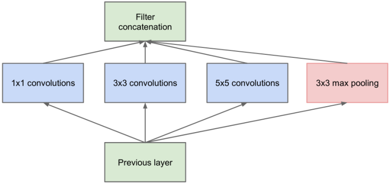


Source: [Szegedy et al., 2015]



Source: [Szegedy et al., 2015]

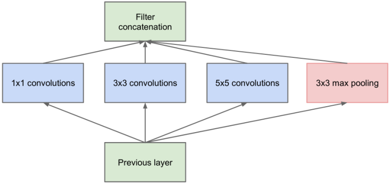
Naïve Inception module



Source: [Szegedy et al., 2015]

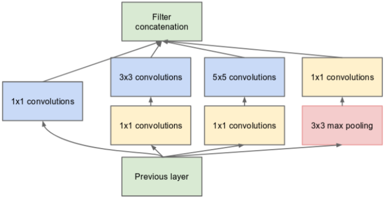
Continuous increase in dimensionality

Naïve Inception module



Source: [Szegedy et al., 2015]

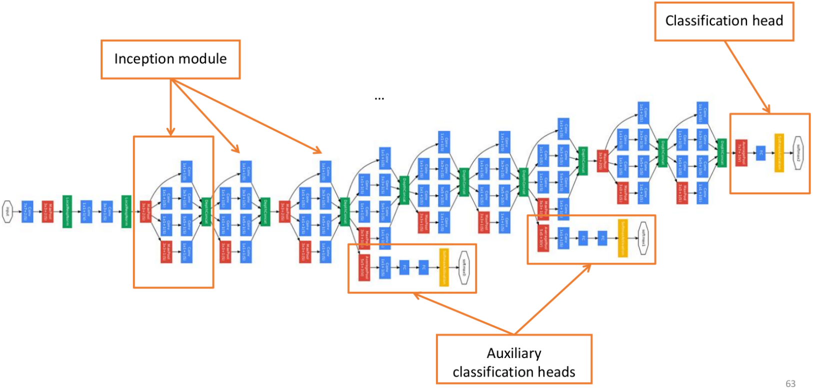
Final Inception module



Source: [Szegedy et al., 2015]

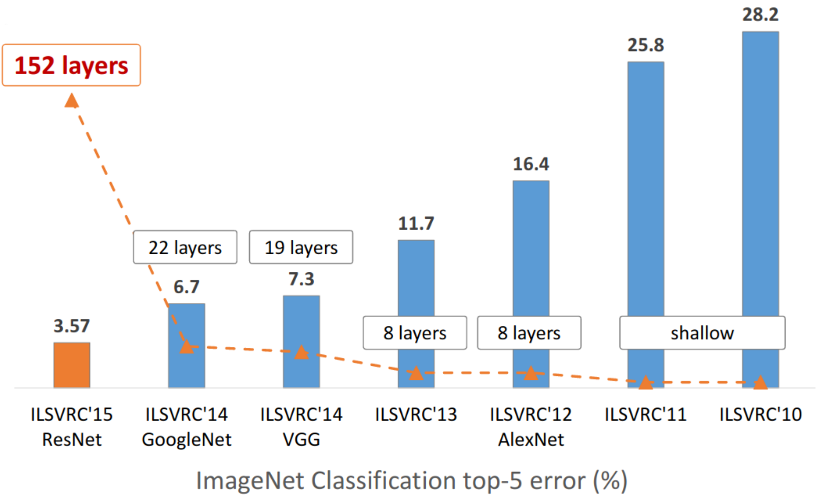
Continuous increase in dimensionality

Dimensionality reduction via 1×1 convolutions (**feature map pooling**)



Source: Otmar Hilliges, ETHZ

- Ensuring sufficient gradient flow through all layers of these networks was challenging
- VGG and GoogleNet used “hacks” to do it:
    - VGG: first a 11-layer model was trained to convergence, then additional random layers were added in the middle
    - GoogleNet: Auxiliary classifiers to inject extra gradient into the lower layers of the network
- Shortly afterwards, **batch normalisation** was invented, removing the need for such hacks



Source: Kaiming He

Microsoft Research

MSRA @ ILSVRC & COCO 2015 Competitions

- **1st places in all five main tracks**
  - ImageNet Classification: “*Ultra-deep*” (quote Yann) 152-layer nets
  - ImageNet Detection: 16% better than 2nd
  - ImageNet Localization: 27% better than 2nd
  - COCO Detection: 11% better than 2nd
  - COCO Segmentation: 12% better than 2nd

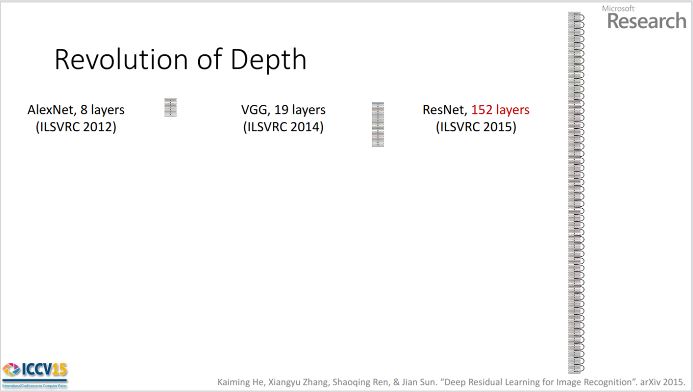
*improvements are relative numbers

ICCV15 International Conference on Computer Vision

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. “Deep Residual Learning for Image Recognition”. arXiv 2015.

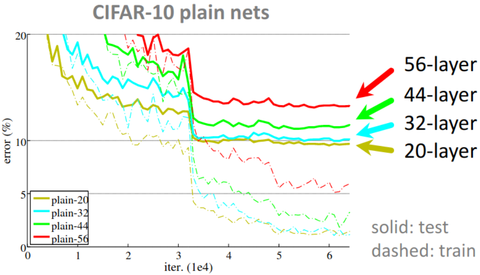
Source: Kaiming He

Slide source: [He et al., 2016]



Source: Kaiming He

- 2-3 weeks of training on a 8-GPU machine
- Faster than a VGG at runtime (although 8x more layers)!

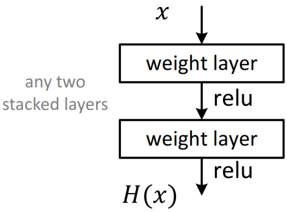


Source: Kaiming He

Failure to optimise

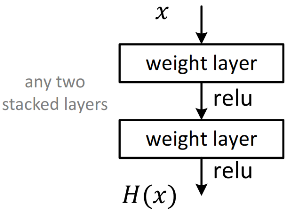
- A deeper network should perform at least as well as a shallow architecture
- "Proof": take a trained shallow network, copy layers and set the output of additional layers to identity

Plain net



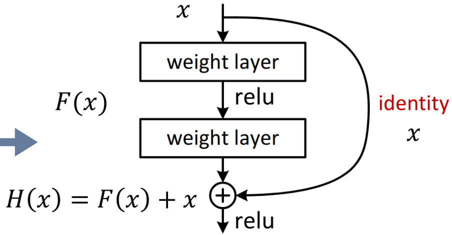
Source: Kaiming He

Plain net



Source: Kaiming He

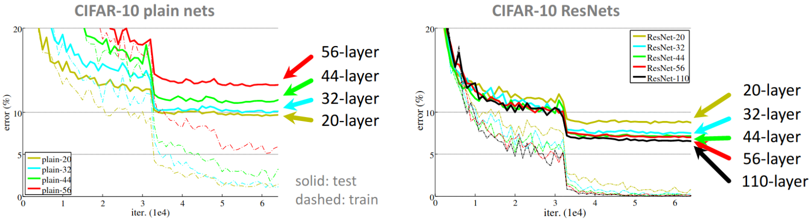
Residual net



Source: Kaiming He

$F(x)$ is a residual mapping w.r.t. identity

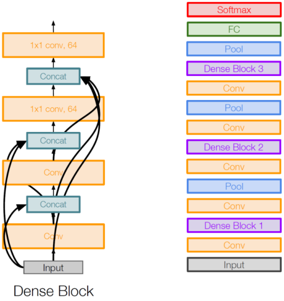
- If identity is optimal, easy to set weights to 0 $\rightarrow$ network can learn to **not use blocks that it doesn't need**
- If optimal mapping is closer to identity, it is easier to **find small fluctuations**
- Residual connections **improve gradient flow**: at add gates in the backward path, the incoming upstream gradient can directly flow through the residual connection



Source: Kaiming He

- Deep ResNets can be trained without difficulties
- Deeper ResNets have lower training error and lower test error
- Further shortcut solutions...

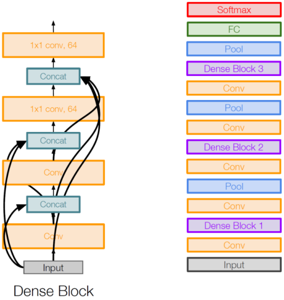
DenseNet



Source: [Huang et al., 2017]

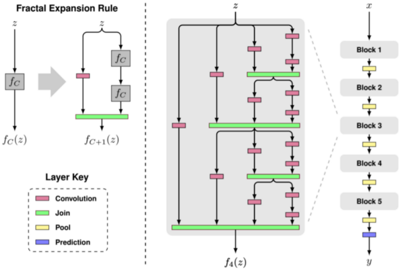
layer l_n receives feature maps of **all** preceding layers $l_0 \ldots l_{n-1}$

DenseNet



Source: [Huang et al., 2017]

FractalNet



Source: [Larsson et al., 2016]

layer l_n receives feature maps of **all** preceding layers $l_0 \dots l_{n-1}$

- Wide ResNet [Zagoruyko and Komodakis, 2016]
- ResNeXt [Xie et al., 2017]
- CondenseNet [Huang et al., 2018]
- ...

And **fully convolutional networks**

- HourGlass [Newell et al., 2016]
- U-Net [Ronneberger et al., 2015]

- Overview of deep CNN topologies
- Convolutional Neural Networks
- Demystifying the Mathematics behind CNNs

Next Lecture

- CNNs in Computer Vision

J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. Imagenet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009.

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S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He. Aggregated residual transformations for deep neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1492–1500, 2017.

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